

Exam Date & Time: 02-Jun-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, MAY-JUNE 2023

LINUX PROGRAMMING [CSE4303]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

- 1) Illustrate the difference between su and sudo with suitable examples. (5)
 - A)
 - B) Write a shell script that accepts a number and check if it is prime or not. (3)
 - C) How Linux differs from other operating systems? (2)
- 2) Explain Linux file permission structure in detail. Write suitable commands to change file permissions with example. (5)
 - A)
 - B) Interpret the purpose of the following vi editor commands.
 - i)d0
 - ii)2yy (3)
 - iii)x
 - C) Demonstrate any two variants of ls command with its functionality. (2)
- 3) Explain the following search pattern that can be used with grep filter (5)
 - i)gg* ii) [^a-zA-Z] iii) [pqr] iv)* v) .* (5)
 - A)
 - B) Explain following terms in git.
 - i)git log (3)
 - ii)git status

- C) Explain the functionality of bin and lib directory of Linux filesystem. (2)
- 4) Assume you have a directory containing a large number of files with names that include spaces and special characters. Your task is to write a script that will rename all the files in the directory to be lowercase and replace spaces with underscores. Write a shell script using a for loop to accomplish this task. (5)
- A)
- B) Answer the following with respect to process handling in Linux:
- i. What command can be used to view a list of currently running processes in Linux? Give an example of how this command might be used.
 - ii. How can the "kill" command be used to terminate a process in Linux? (3)
 - iii. What command can be used to change the priority of a running process in Linux? Give an example.
- C) What is the Linux kernel, and what is its role in the operating system? (2)
- 5) Write AWK program for the following tasks:
- A)
- i. To count the number of occurrences of each word in a file, using an associative array to keep track of the counts and a loop to iterate over the words. (5)
 - ii. To print the lines of a file that contain a pattern "your_name" repeated more than three times. Use a loop to iterate over the characters of the pattern.
- B)
- Given a text file "example.txt" containing some lines, use SED editor to perform the following operations:
- i. Delete the first line.
 - ii. Replace all occurrences of the word "apple" with "orange". (3)
 - iii. Append the text "The end" to the end of the file.
- C) Evaluate the importance of the "svn log" command in Subversion Control. (2)

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